Class Meeting Handout #5

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Today we end our review with an investigation into the unknowable structure of transition states.

Transition States and Energy Surfaces

You will have discussed transition state theory in detail in your 2nd-year physical chemistry course. Let us see if we can come at it from a different angle, with the needs of physical organic chemistry in mind. Today we will review reaction coordinates, transition states, and the Hammond postulate.[1]

[1] You will have encountered these ideas as you read chapters 7.1, 7.2 & 7.3 of the textbook in preparation for this class. The Moodle site will describe the required and suggested reading for *Before* and *After* each class meeting.

Review: Transition States – 20 minutes

1. Draw the mechanism for the S_N2 substitution of benzyl bromide using a bromide nucleophile. The starting material and product will be identical, but the bromide atom is exchanged. We will be using acetone as the solvent.[2] Use curved arrows to show electron flow. Draw a reaction coordinate energy diagram representing this reaction. Justify your relative energy the initial and final condition.
2. Draw the transition state. Are both carbon to bromine associations in the transition state identical?
3. Draw the mechanism for the S_N2 substitution of benzyl bromide using a chloride nucleophile. Draw the transition state and predict the relative bonding between the incoming nucleophile and the carbon atom and the carbon atom and the leaving group. Are they identical? Or not? Explain your answer. Draw the reaction coordinate. Is the transition state at the same location (exactly "50%")?[3]
4. Do all that again using iodide as the nucleophile.

[2] Recall the effect of solvent on the nucleophilicity of halogen ions. In polar aprotic solvents, such as acetone, chloride is more nucleophilic than iodide because of its more concentrated charge. In protic solvents, such as methanol, that order is reversed because the chloride ion is better solvated, This trend is not a rule and the truth is complex. McMurray's *Organic Chemistry* avoids this topic, perhaps he is aware that stating these trends as rules can set students up for difficulty in their 4th-year physical organic course. Our textbook has a discussion on this issue in chapter 8.4.4 and we will be getting to that later in this course.

[3] How does the Hammond Postulate apply here?

Exercise: Changing Transition State Structure – 20 minutes

Is there a better way to express changing transition state structure in the reactions above?

5. What creates the values used in the x-axis of the reaction coordinate diagrams you made above?
6. Draw a plot with the distance between the incoming nucleophile and the carbon atom on the x-axis and the distance from the carbon atom and the leaving group on the y-axis.

7. Trace the three reaction coordinates that you presented above on this plot. Locate the transition state in each case.

8. Now apply the Hammond effect (along the reaction coordinate) and the anti-Hammond effect (perpendicular to the reaction coordinate) to evaluate and correct your hypotheses.

   You have just completed your first More O'Ferrall-Jencks plot.[4]

[4] You will have encountered the idea of More O'Ferrall-Jencks plots as you read chapter 7.8 of the textbook in preparation for this class.

Discussion: Assignment Topics – 10 minutes

We will discuss the problems from Appendix 5 of the textbook that were the focus of assignment #1. The assignment was due before this class meeting so I expect that we are all prepared.[5]

[5] There are more problems in appendix 5 of your textbook that you should practice. Hoffman's *Organic Chemistry: an Intermediate text* has more good practice problems for reviewing organic mechanism, especially chapters 4, 7 & 8.

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

1. Draw a mechanism using curved arrows and draw the transition state.

2. Define and apply the Hammond postulate.

3. Predict changes in T.S. structure as we change the structure of the reactants.

4. Understand that changes in bonds as we pass through a transition state can be concurrent but asynchronous and how this reflects in the structure of the T.S. and the path of the reaction coordinate.

Looking Ahead

We have reviewed how to hypothesize the structure of a transition state and used the Hammond postulate to predict how that structure might change as reactants (or conditions) change. We have introduced a new way to express these ideas, the More O'Ferral-Jencks plot. A major goal of this course is to explore methods for probing transition state structure. We will be using the More O'Ferral-Jencks plot many more times.

The next assignment will focus on the postulates and axioms that underpin the field of physical organic chemistry. We have had a start here with the Hammond postulate.[6]

Our review is now complete.[7] Next we move on to the subject of acid-base chemistry. It will start with review, but there is more to acids and bases than you have seen so far.[8] We will make complex rate laws for reactions that involve proton transfer. We will be both general and specific.[9] What is the pH of 100% sulphuric acid?[10] You are now ready for the truth.

[6] After completing assignment #2, you will also be better acquainted with the reactivity-selectivity principle, the principle of microscopic reversibility, the Curtin-Hammett principle and the concept of thermodynamic and kinetic control.

[7] Really? There seems to be so much more we could have talked about.

[8] I knew we weren't done with review. This course applies almost everything you have learned in chemistry so far. We will pause to briefly review whenever needed.

[9] Did you see what I did there?

[10] How do we express a value for protonating power (pH is one such description) when we are not in water anymore? Hammett has an answer.